Sealed-seam suppression in Froggatt–Nielsen-type models: two theorems, an obstruction by conjunction, and a discrete neutrino test

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preprint, not peer reviewed, version 2026-06-11 (substantially rebuilt after hostile review; the withdrawn claims of the earlier draft are listed in Section 7 and stay withdrawn).

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Abstract

Froggatt–Nielsen (FN) mechanisms generate fermion mass hierarchies as powers of a small parameter λ, one insertion per unit of flavor charge, with λ fitted. We study an FN-type setting in which the suppression is *pinned by a theorem*: for sealed two-outcome records — rank-one POVM-type letter pairs with weights p and $1-p$ in a stationary representation — (i) the transition norm is $\sqrt{p(1-p)}\,|\langle u, v\rangle| \le \sqrt{p(1-p)}$, and (ii) a misaligned sealed pair cannot support a stationary state (top transfer eigenvalue $\frac{1}{2}[1+\sqrt{1-4p(1-p)(1-|c|^2)}] < 1$ strictly), so stationarity forces saturation: $\|S\| = \sqrt{p(1-p)}$ exactly, with the seam phase pure rephasing gauge. An exact selection rule follows in any algebra graded by a family charge whose only charged sealed generator is the seam: amplitudes vanish unless the seam count balances the charge difference (receipts: 0/27,000 violations; bounds saturated to ten digits). The paper's central *negative* result is an **obstruction by conjunction**, found when the hostile review of an earlier draft demolished its seesaw realization and we completed the demolition: for *integer* right-handed charges, Majorana diagonals carry even powers of λ and an exhaustive Dirac-consistent scan over rule-allowed textures (charges in $\{0..3\}^3$, unit $\pm$ coefficients) never reaches the half-integer-rung spectrum (zero exact hits; best deviation 0.014; receipted); for *half-odd* right-handed charges the Majorana side can reach the rungs exactly, but the corresponding Dirac column carries half-integer charge, balances no integer seam count, and vanishes identically — rank collapse; and the textbook cancellation in any case removes RH charges from the light masses when the Dirac sector is charged consistently. The surviving realization places half-unit charges on the lepton doublets (Weinberg-operator dressing), where the same selection rule yields a **discrete menu** for the light spectrum at unit coefficients (four full-rank members; two already excluded by data): the rung point $S = \sqrt{\varepsilon_{\text{eff}}/(1+\varepsilon_{\text{eff}})} = 0.172747$ (three insertions) and the cheaper texture $S = 0.167880$ (two insertions), with $\varepsilon_{\text{eff}} = \varepsilon(1-\varepsilon)$ and $\varepsilon = 0.031768636446582$ a constant whose self-contained pedigree is the mathematical companion [C0]. Against JUNO's first data ($S = 0.17283 \pm 0.00167$, our combination of JUNO's Δm^2_{21} with the global-fit Δm^2_{31}) the data-live members sit at -0.05σ and -3.0σ; they are separated by 7.3σ at JUNO-final precision,

so the menu is adjudicable this decade, and the minimality principle the program uses elsewhere favors — on current data — the *worse* member, which we state rather than hide. What this paper does **not** deliver, withdrawn from an earlier draft and listed with reasons: mixing-angle and CP predictions (they rested on an unlisted measure-zero orthogonality assumption), the claim that the mechanism's flavor symmetry uniquely "predicts its own mediator," and any charged-fermion statement. One quantitative input is irreducible and stated in one line: the seam weight is the companion theory's first-order marginality, $p = \varepsilon$ — a referent scan shows it underivable from the obvious fixed-point quantities, and it carries the aggravating fact that the first-order law it comes from has its sign overturned at second order at the defining weight. The look-elsewhere cost of the dressed value is counted (six of twelve natural one-factor forms fit today's central value; three are mutually inseparable even at JUNO-final), and the claim chronology relative to JUNO's first release is disclosed in full.

1. The Seam Theorem

Work in the representation every reflection-positive stationary process admits (companion [C2], Thm 2.1): Hilbert space, PSD letters, $T = \sum_x A_x \preceq \mathbf{1}$, stationary unit vector $T\omega = \omega$.

Definition (S-atom premise, named). A *sealed two-outcome seam* is a letter pair of rank one: $A_0 = p\,uu^\dagger$, $A_1 = (1-p)\,vv^\dagger$, unit u, v.

Lemma 1 (norm). $\sqrt{A_0}\sqrt{A_1} = \sqrt{p(1-p)}\,\langle u, v\rangle\, uv^\dagger$, so $\|\sqrt{A_0}\sqrt{A_1}\| = \sqrt{p(1-p)}\,|\langle u, v\rangle| \le \sqrt{p(1-p)}$, with equality iff the poles align.

Lemma 2 (stationarity gate). On the seam span, with $c = \langle u, v\rangle$ and $d = 4p(1-p)(1-|c|^2)$: $\lambda_{\max}(A_0 + A_1) = \frac{1}{2}[1 + \sqrt{1-d}]$, equal to 1 iff $|c| = 1$; the exact gap is $\frac{1}{2}[1 - \sqrt{1-d}] \ge p(1-p)(1-|c|^2)/2$, strictly positive for misaligned poles. (An earlier draft's abstract quoted the lower bound as the exact gap; corrected.)

Theorem 1 (stationarity forces saturation). If the seam block acts invariantly at stationarity (premise: decoupling) and the stationary state overlaps it, then alignment is forced and the transition norm is exactly $\sqrt{p(1-p)}$. At alignment the letters commute: the extremal seam is classical, and the factor is the two-pole weight of a definite record. The seam phase ($u \mapsto e^{i\varphi}u$) changes no observable — the rephasing structure mass terms have always had.

Receipts: norm identity to 2.8×10^{-16} (200 random complex seams, $d = 2..5$); eigenvalue closed form to 3.3×10^{-16}.

2. The selection rule

Theorem 2. In an algebra carrying a graded charge Q with every sealed generator Q-neutral except one seam S shifting Q by one unit, $\langle x'|W|x\rangle = 0$ for every word whose seam count fails to balance $x' - x$ (grade decomposition), and — since neutral letters are contractions — balanced amplitudes obey $|\langle x'|W|x\rangle| \le \|S\|^{|x'-x|}$ (submultiplicativity; an inequality, with saturation a property of the specific model, exhibited in the receipts).

Receipts: 0 violations in 27,000 matrix elements; maximal one- and two-unit amplitudes 0.175384 and 0.030759, saturating $\sqrt{\varepsilon_{\text{eff}}}$ and ε_{eff} to ten digits in the explicit model.

Scope correction (withdrawn corollary). An earlier draft claimed ”oscillations require the seam — the mechanism predicts its own mediator.” That conflated two gradings: if the Dirac sector is charged consistently, the claim’s premise breaks the spectrum (Section 3); if the Dirac sector is outside the graded algebra, sectors connect without the seam and the corollary is vacuous. Withdrawn. What survives is the unglamorous statement: *within the graded (sealed) sector*, charge steps cost seam insertions.

3. The obstruction by conjunction (central result)

The natural realization — FN charges on the right-handed seals, M_R hierarchical, Dirac sector neutral — **fails twice over**, and we record both failures as the paper’s main content:

(a) The textbook cancellation. If the Dirac couplings are charged consistently with the grading, the FN suppressions cancel identically in the seesaw, $m_\nu = v^2\, Y M_R^{-1} Y^\mathsf{T}$ — the well-known result that FN charges on singlet neutrinos drop out of the light masses (see e.g. [2]); and half-integer RH charges make the Dirac column amplitudes exactly zero under a unit-shifting seam, collapsing the rank. An earlier draft evaded this by declaring the Dirac sector ”neutral” while keeping the selection rule — an inconsistency the review exposed.

(b) Parity, scoped, plus the scan. Majorana bilinears $\nu_i^c \nu_j^c$ carry total charge $x_i + x_j$. For **integer** charges the diagonals carry even powers of λ — a parity statement that by itself constrains only diagonal-dominant textures (off-diagonal-dominant blocks can carry odd-power eigenvalue magnitudes), so the integer-charge obstruction is completed by *exhaustive enumeration*: over charge vectors in $\{0..3\}^3$ with unit $\pm$ coefficients on every rule-allowed entry, **zero** textures reach the rung spectrum. The closest approach — metric pinned as the max over the two independent ratio deviations $|m_1/m_3 - \varepsilon_{\text{eff}}|$ and $|m_2/m_3 - \sqrt{\varepsilon_{\text{eff}}}|$, **computed on the seesaw-inverted (light) spectrum**: the charges sit on M_R and the rung target is for the light masses, so $m_i \propto 1/\mu_i$ with μ_i the eigenvalue magnitudes of the scanned texture (equivalently, the two components are $|\mu_1/\mu_3 - \varepsilon_{\text{eff}}|$ and $|\mu_1/\mu_2 - \sqrt{\varepsilon_{\text{eff}}}|$) — is 0.014, with the **witness published** so the figure is checkable from one matrix: $q = (3, 2, 3)$, coefficient pattern $\{11{:}{+1}, 12{:}{+1}, 13{:}{+1}, 22{:}{-1}, 23{:}{+1}, 33{:}0\}$ on magnitudes $\lambda^{q_i+q_j}$, component deviations $(0.0144, 0.0092)$. (Applying the metric to the heavy spectrum *uninverted* gives $(0.0144, 0.0770)$ on this same witness; an earlier draft attributed that 0.077 to ”a less complete scan variant” — it is in fact the witness’s own uninverted second component, and the inversion convention was unstated — the same class of convention bug a previous revision fixed for the charge box, caught in review and now pinned.) The zero-exact-hits statement is robust under the pinned convention: a full independent rerun of the $\{0..3\}^3 \times \{0, \pm 1\}^6$ scan confirms zero exact hits, best deviation 0.0144. Scope honesty: the charge box $\{0..3\}^3$ is a choice; entries beyond it carry magnitudes $\leq \lambda^7 \approx 5 \times 10^{-6}$ and push textures further from the target, but ”exhaustive” means exhaustive-within-the-box. For **half-odd** charges, parity does *not* obstruct the Majorana side — $q = (1, \frac{1}{2}, 0)$ makes $M_R = \text{diag}(\lambda^2, \lambda, 1)$ rule-allowed and exactly on the rungs — but clause (a) kills the realization: the $q = \frac{1}{2}$ Dirac column carries half-integer charge, balances no integer seam count, and is exactly zero, collapsing the light-mass rank. **The obstruction is the conjunction of (a) and (b)-plus-scan**, and we state it as such; an earlier draft presented (b) as a standalone parity theorem (false for half the charge lattice) and left the scan’s spectrum-inversion convention unstated (the source of the misattributed 0.077 above) — both corrected here, with the scan scope and conventions now pinned.

The surviving realization. Half-unit charges on the lepton *doublets*, $q_L = (1, \frac{1}{2}, 0)$, acting on the Weinberg operator $(L_iH)(L_jH)$ with entry charges q_i+q_j and magnitudes $\lambda^{q_i+q_j}$ at integer insertions ($\lambda = \sqrt{\varepsilon_{\rm eff}}$): allowed entries are $(11), (22), (33), (13)$; (12) and (23) are parity-forbidden. This realization is *selected, not chosen*, by two facts: (i) **integer** doublet charges on the Weinberg operator are excluded by the same exhaustive enumeration run on the *direct* spectrum (no seesaw inversion here) — zero exact hits over the full $\{0..3\}^3 \times \{0, \pm1\}^6$ box, 30,624 full-rank nondegenerate textures, best deviation 0.0770, witness in the receipts; and (ii) $q_L = (1, \frac{1}{2}, 0)$ is the *unique* normalized assignment realizing the rung diagonal: $\mathrm{diag}(\lambda^{2q_1}, \lambda^{2q_2}, \lambda^{2q_3}) = \mathrm{diag}(\lambda^2, \lambda, 1)$ forces $2q_1 = 2$, $2q_2 = 1$, with the common shift fixed by $q_3 = 0$ (a shift rescales the spectrum, not its ratios). Convention, pinned: entries are 0 or ±1 times the allowed magnitude; full rank and a nondegenerate ordering ($m_1 < m_2 < m_3$, all nonzero) required. The exhaustive menu then has exactly **four** members (a previous revision said three, missing the quasi-degenerate fourth — caught in review; for a paper whose method is exhaustive enumeration, we print the complete list):

```
texture             insertions  spectrum ratios       S
diag(l^2, l, 1)         3       (e_eff, sqrt(e_eff), 1) 0.172747
m13 + m22 + m33         2       (0.0290, 0.1703, 1)     0.167880
all four, sgn(m11
 m33) = -1              4       (0.0581, 0.1704, 1)     0.160500
m11 + m22 + m13,
 m33 = 0                4       (0.839, 0.916, 1)       0.675516
(the third and fourth are already excluded by data - at 7.4
 and ~300 sigma respectively - and are listed as falsified
 members, not hidden; an earlier draft printed 0.169993 for
 the second member, a wrong closed-form shortcut - the correct
 value follows from the printed ratios themselves: S^2 =
 m1/(1+2m1), m1 = (sqrt(1+4*e_eff)-1)/2, receipted)
```

4. The two-point test

Against JUNO's first data — $S = 0.17283 \pm 0.00167$, **our combination** of JUNO's $\Delta m^2_{21} = (7.50 \pm 0.12) \times 10^{-5}$ [3] with the global-fit $\Delta m^2_{31} = (2.511 \pm 0.027) \times 10^{-3}$ [6], not a JUNO-only number:

```
member              S           now        notes
rung point          0.172747    -0.05 s    (reference)
cheap texture       0.167880    -3.0  s    7.3 s from the rung
                                           point at JUNO-final
third member        0.160500    -7.4  s    already excluded;
fourth member       0.675516    excluded   listed, not hidden
(undressed point    0.175472    +1.6  s    the registered claim of
 for comparison                            the companion note [VII])
```

The two data-live members are separated by 7.3σ at JUNO-final precision ($\sigma_S^{\rm final} \approx 0.00067$: the projected sub-percent splittings, $\sigma_{21} \approx 0.6\%$ and $\sigma_{31} \approx 0.5\%$ [5], propagated as half the quadrature sum of the relative errors times S): the menu is adjudicable this decade. Preassigned outcome bands at JUNO-final, all branches covered: data within 2σ of 0.172747 — rung point survives, cheap texture dead; within 2σ of 0.167880 — the reverse, and minimality scores a sharp win; **within $2\sigma_{\rm final}$ of 0.175472 — both menu members die while [VII]'s**

undressed point survives; anywhere else — the whole construction is dead and we say so. Stated against interest: the program's own minimality principle (fewest insertions), applied as elsewhere in the corpus, favors the **cheap texture** — currently the *worst-fitting live member at* -3.0σ. If JUNO-final confirms today's central value, minimality-as-selector fails here and we will say so.

What the menu does *not* include: mixing angles and CP phases. The allowed Weinberg textures are too sparse to generate the observed mixing from the neutrino sector alone (the $(12), (23)$ entries are parity-forbidden), so the PMNS matrix must come dominantly from the charged-lepton sector — the earlier draft's θ_{23} and δ_{CP} "predictions" are **withdrawn** (Section 7). One genuine charged-sector *constraint* the realization does impose, stated since review caught us denying it: $q_{L_2} = \frac{1}{2}$ makes the muon Yukawa row $\bar{L}_2 e_R H$ carry half-integer charge, which vanishes unless the right-handed charged-lepton charges compensate modulo 1 — half-odd e_{R_2} is an **additional named assumption** (a massless muon is not negotiable). So the mechanism says nothing about charged *masses* but does constrain charged *charges*; the abstract's withdrawal item is scoped accordingly.

5. The one-line input, with its aggravating fact

The mechanism's single quantitative input: the seam weight is the companion theory's first-order marginality, $p = \varepsilon = 3\kappa - 1$, with κ in closed form ([C0]; no Standard-Model input anywhere in the chain). A pre-registered referent scan (ten exact fixed-point candidates, 25 digits) found no independent quantity equal to ε: the identification is not derived, and we state the aggravating fact a referee located: ε is the coefficient of a *first-order* law whose sign prediction fails at the defining weight $w = 3$ (first-order -0.0051 vs exact $+0.0084$; [C0] itself warns against extrapolating its truncations). The input is therefore: *the seam weight is a specific coefficient of an expansion that is not trustworthy at the weight that motivates it.* We keep the input because the resulting point sits at -0.05σ and the falsification machinery is cheap; we decline to dress it up.

6. Chronology, trials, and look-elsewhere

Disclosed in full. (i) The *value* $\varepsilon_{\text{eff}} = \varepsilon(1-\varepsilon)$ was noticed **after** JUNO's first release; the program's registered claim is the *undressed* point (0.175472 at exact ε; see [VII] and its precision correction), which that data disfavors at 1.6σ. (ii) The look-elsewhere cost was counted before this mechanism existed. The twelve forms, enumerated (base $\varepsilon \cdot f$): $f \in \{1-\varepsilon,\ 1/(1+\varepsilon),\ e^{-\varepsilon},\ (1-\varepsilon)^2,\ 1-2\varepsilon,\ 1-\varepsilon/2,\ 1/(1+2\varepsilon),\ 1-\varepsilon^2,\ \cos\varepsilon,\ 1-\kappa\varepsilon,\ 1-\theta\varepsilon,\ 1-\eta\varepsilon\}$: six fit the current central value within 1σ; three (including $\varepsilon_{\text{eff}} = \varepsilon(1-\varepsilon)$) are mutually inseparable at 0.2σ even at JUNO-final — so the data can confirm or kill the one-factor *class*, never select $\varepsilon(1-\varepsilon)$ within it, and a theorem pinning any cluster member would have looked equally prophetic. The Seam Theorem's evidential weight is therefore *zero until its premises are tested elsewhere*; its value is that it converts a free parameter into a falsifiable conjunction. (iii) No claim in this paper is "registered" in any externally verifiable sense until the program's named deposit exists; the internal registration documents carry their own dating and say the same.

7. Withdrawn claims (kept visible)

1. "$\sin^2\theta_{23} = 1/2$ exactly" — rested on requiring an *orthogonal* neutral Dirac matrix, a measure-zero assumption with no symmetry behind it, which also smuggled CP conservation (real orthogonal) and silently fixed θ_{12}, θ_{13} to excluded values. Withdrawn as a mechanism prediction; what remains is a texture observation in the program's corpus (its registration document carries the same withdrawal).

2. "$J = 0$, $\delta_{CP} \in \{0, \pi\}$" — the phase-gauge argument covers the seam (Majorana) sector only; the Dirac and charged-lepton phases are untouched. Withdrawn.

3. "Oscillations require the seam" — grading-scope conflation (Section 2). Withdrawn.

4. The discrete $m_{\beta\beta}$ menu — computed with undisclosed measured mixing angles as inputs; meaningless once (1) is withdrawn. Withdrawn.

5. "Exact rungs for any neutral Dirac sector" — exactness was an artifact of the orthogonality fiat; the honest exact statement is the Weinberg-realization menu of Section 3.

6. All charged-fermion *mass* statements — the earlier draft quantified charged-sector "gaps" with an unreproducible metric; no such number survives. (Not withdrawn: the charge-mod-1 compensation *constraint* of Section 4, which the realization genuinely imposes and which carries its own named assumption.)

8. Relation to Froggatt–Nielsen models

The mechanism class is FN [1]. Prior art on *derived* suppression exists and is engaged per our standing commitment: in anomalous- $U(1)$ (Green–Schwarz) FN models the parameter is computed from the Fayet–Iliopoulos term, $\lambda \approx 0.17$–0.22 (Binétruy–Ramond; Dudas–Pokorski–Savoy; Ibáñez–Ross [4]) — a range that, we note with appropriate discomfort, contains $\sqrt{\varepsilon_{\text{eff}}} = 0.175$. What is different here: the suppression is pinned by a stationarity theorem in a record formalism rather than by an anomaly coefficient, the charge's selection rule is exact rather than order-of-magnitude, and the realization question (this paper's obstruction and menu) is answered by exhaustive enumeration at unit coefficients. What is *not* different: like every FN model, the O(1)-coefficient freedom is the mechanism's soft underbelly, and our unit- coefficient discipline is a named choice, not a derivation.

Reproducibility

Fixed-seed scripts regenerate every number (the seam lemmas, the selection-rule receipts, the parity-obstruction scan, the Weinberg menu, the prediction table); bit-identical reruns; code and canonical outputs accompany the submission.

References

[1] C. D. Froggatt, H. B. Nielsen, *Nucl. Phys. B* **147** (1979) 277.

[2] G. Altarelli, F. Feruglio, Models of neutrino masses and mixings, *New J. Phys.* **6** (2004) 106 — including the cancellation of singlet-neutrino FN charges in the seesaw.

[3] JUNO collaboration, arXiv:2511.14593; *Nature* **654** (2026) 343 — first measurement, $\Delta m^2_{21} = (7.50 \pm 0.12) \times 10^{-5}\,\mathrm{eV}^2$.

[4] P. Binétruy, P. Ramond, *Phys. Lett. B* **350** (1995) 49; E. Dudas, S. Pokorski, C. A. Savoy, *Phys. Lett. B* **356** (1995) 45; L. E. Ibáñez, G. G. Ross, *Phys. Lett. B* **332** (1994) 100 — anomalous-$U(1)$ FN with FI-derived λ.

[5] JUNO collaboration, JUNO physics and detector, *Prog. Part. Nucl. Phys.* **123** (2022) 103927 — projected sub-percent splitting precisions (an earlier draft cited the DUNE TDR and Hyper-Kamiokande design report here; those experiments do not measure Δm^2_{21} at sub-percent — misattribution, corrected).

[6] I. Esteban et al., NuFIT — global-fit Δm^2_{31} input; version sensitivity applies to the hybrid S above.

Companions: [C0] the fixed-point theory (the constant's pedigree); [C2] the representation theorem; [VII] the registered undressed point and its correction record (the two claims are mutually exclusive and cross-declared).